ModelIt!

Teacher's Guide

How Do Ecosystems Recover from Catastrophe?

Modeling Ecological Succession, Resilience Thresholds, and Regime Shifts

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-LS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-6

Evaluate the claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students evaluate evidence about ecosystem recovery trajectories, resilience thresholds, and regime shifts to construct arguments about whether specific restoration strategies will achieve desired recovery outcomes.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Students model how the complex interactions among species, nutrient cycles, and physical environment maintain ecosystem stability under normal conditions and drive recovery or transformation after catastrophic disturbance.
Crosscutting Concept	**Stability and Change** — Students identify the feedback mechanisms that maintain ecosystem stability (negative feedbacks in mature systems) and the tipping points where positive feedbacks drive rapid transition to alternative states (regime shifts).

CAST Testing Alignment

MULTIPLE CHOICE:

Two forest plots experienced identical wildfire severity. Plot A has intact soil with a rich seed bank; Plot B has severely damaged soil with no seed bank. Based on the succession model, what is the predicted difference in Recovery Completeness Index after 20 years?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why climate change is lowering ecosystem resilience thresholds worldwide, making regime shifts more likely after disturbances that historically allowed recovery. Reference at least three model components and the concept of alternative stable states.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Ecosystems are not static — they are dynamic systems maintained by feedback loops among species, nutrients, physical environment, and disturbance regimes. When catastrophic disturbance disrupts these feedback loops, the ecosystem must rebuild from whatever biological legacy survives. The science of ecological succession reveals predictable patterns in this rebuilding process, but also shows that recovery is not guaranteed — certain combinations of disturbance severity, changed climate, and biological loss can push ecosystems past tipping points into entirely new configurations.

Primary and Secondary Succession

The distinction between primary and secondary succession is fundamental. Primary succession occurs on bare substrate where no soil exists — newly formed volcanic rock, retreating glacier surfaces, or newly formed sand dunes. Recovery begins with weathering and colonization by lichens and cyanobacteria that slowly create the first millimeters of soil over decades. Secondary succession occurs where soil and biological legacy (seed banks, root networks, soil microbiome) survive — after fire, logging, or storm damage. Because soil formation is the slowest step in succession, the presence of even degraded soil can accelerate recovery by decades or centuries. Mount St. Helens provided a natural experiment: areas where volcanic ash buried but did not destroy soil recovered 10-50x faster than areas where pyroclastic flows sterilized the substrate entirely.

Resilience and Regime Shifts

Ecological resilience is not infinite — every ecosystem has thresholds beyond which recovery to the original state becomes impossible. When these thresholds are crossed, the ecosystem undergoes a regime shift to an alternative stable state maintained by different feedback loops. Classic examples include: coral reefs shifting to algae-dominated systems when overfishing removes herbivorous fish, eutrophic lakes flipping from clear to permanently turbid states when phosphorus loading exceeds a threshold, and boreal forests converting to grassland when fire frequency increases beyond the rate at which trees can regenerate. Once a regime shift occurs, reversing it requires not just removing the original stressor but overcoming the new feedback loops that maintain the alternative state — which may require much larger intervention than preventing the shift would have required.

Climate Change and Novel Ecosystems

Climate change is fundamentally altering the conditions under which ecosystems must recover from disturbance. A forest that burned in 1920 could regrow the same species because climate conditions were similar to pre-fire conditions. A forest that burns today may face 1-3 degrees Celsius warming and altered precipitation that no longer support the original species. This creates novel ecosystems — species combinations that have no historical analog, assembled from whatever species can tolerate current conditions regardless of whether they historically co-occurred. Restoration ecologists face a fundamental choice: attempt to recreate the historical ecosystem (which may fail under changed climate) or facilitate adaptation to projected future conditions (which means accepting the loss of the original ecosystem).

Invasive Species and Successional Disruption

Disturbed ecosystems are exceptionally vulnerable to invasive species because disturbance creates exactly the conditions invasives thrive in: open space, available light, reduced competition, and exposed soil nutrients. In western North America, cheatgrass (Bromus tectorum) invades burned areas, grows faster than native species, and cures into highly flammable dry fuel that increases fire frequency from the historical 30-100 year interval to 3-5 years. This fire-invasion feedback loop permanently converts sagebrush steppe and juniper woodland to annual grassland — a regime shift driven by biological invasion rather than physical disturbance. Managing invasive species during the vulnerability window of early succession is critical for enabling native community recovery.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the ecosystem recovery system: disturbance severity, seed bank availability, climate suitability, pioneer establishment, soil development, biodiversity accumulation, nutrient cycling, trophic web complexity, invasive species pressure, and recovery completeness.

E **Establish Relationships**

Students map relationships showing how disturbance severity and seed bank availability determine initial recovery speed, how soil development constrains all subsequent succession, and how invasive species and climate change can divert recovery toward alternative states.

V **Visualize & Model**

Students build a ten-component computational model showing the successional trajectory with feedback loops between soil development, biodiversity, nutrient cycling, and trophic complexity, and threshold effects where invasive species or climate change trigger regime shifts.

E **Evaluate Outcomes**

Students run post-fire secondary succession, post-volcanic primary succession, and climate-shifted recovery scenarios to identify the critical variables that determine whether an ecosystem recovers, transforms, or collapses.

R **Revise & Extend**

Students add mycorrhizal network recovery, keystone species reintroduction, or legacy effect strength to model how underground networks, ecological engineers, and biological inheritance influence successional trajectory.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"On May 18, 1980, Mount St. Helens erupted with the force of 27,000 nuclear bombs, leveling 230 square miles of forest in minutes. Scientists said it would take centuries to recover. They were wrong about some areas — and right about others. Why? The answer reveals how ecosystems rebuild and what can stop them."*

Watch For

Show cover image with before/after photos of Mount St. Helens. Ask: what determined which areas recovered quickly and which are still barren 45 years later? Catalog student hypotheses.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling how ecosystems rebuild from catastrophe — and discovering that recovery is not guaranteed. You will learn what determines whether a forest regrows, converts to grassland, or becomes something entirely new."*

Watch For

Have students read objectives. Pre-teach 'regime shift' with a ball-and-cup analogy: a ball in a cup (stable state) can withstand small pushes, but a large enough push sends it over the rim into a different cup (alternative stable state). Going back requires pushing over the rim again.

Slide 3: Big Question

3 min

Say: *"After catastrophe, an ecosystem does not just bounce back — it rebuilds through a complex sequence that can take decades to centuries. And sometimes it does not rebuild at all. What makes the difference between recovery and permanent transformation?"*

Watch For

Quick-write: Students list what they think an ecosystem needs to recover from a wildfire. Share out and discuss: which factors are most important and why? Are any of them missing after certain types of disturbance?

Slide 4: LEVER Framework

2 min

Say: *"We will Label the ten components that control ecosystem recovery, Establish how they interact through feedback loops, Visualize the successional trajectory, Evaluate three different recovery scenarios, and Refine by adding underground networks and keystone species."*

Watch For

Preview each LEVER step. Emphasize that this model will explain not just how ecosystems recover but what determines whether they recover at all — the concept of resilience thresholds.

Slide 5: Activity 1: Components

8 min

Say: *"Three inputs are determined by the disturbance and environment: how bad the damage is, what seeds survived, and whether climate still supports the original ecosystem. Seven represent the recovery processes that must occur in sequence. Sort them and identify which processes are bottlenecks."*

Watch For

Students sort external versus internal components. Discuss: why is Climate Suitability external? Because climate change has shifted the baseline conditions under which recovery must occur, and the recovering ecosystem cannot change the climate.

Slide 6: Activity 2: Connections

8 min

Say: *"Pioneer species modify soil. Soil supports more species. More species create complex food webs. Complex food webs stabilize nutrient cycles. But invasive species can intercept this chain at any point. Trace the recovery pathway and identify where it can be disrupted."*

Watch For

Students draw arrows showing the successional cascade. Identify the invasive species disruption points and the climate suitability constraint that determines whether the original endpoint is achievable.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run three scenarios: post-fire secondary succession with intact soil, primary succession from bare volcanic rock, and recovery under changed climate. Watch the Recovery Completeness Index over time and identify when and why the trajectories diverge."*

Watch For

Students run all three scenarios and overlay the recovery curves. Key discussion: the climate-shifted scenario never reaches 100% — why? Because the target ecosystem has changed. Is this failure or adaptation?

Slide 8: Discoveries

5 min

Say: *"Your model reveals that soil is the master bottleneck, seed banks determine initial trajectory, resilience thresholds are real, invasive species can permanently redirect succession, and climate change is lowering resilience*

worldwide. Recovery is possible — but not inevitable."

Watch For

Class discussion of discoveries. Ask: in a world of increasing disturbance frequency and changing climate, should restoration ecology aim to recreate the past or facilitate adaptation to the future?

Slide 9: STEM Challenge

12 min

Say: *"50,000 acres of Sierra Nevada forest have burned. Soil is damaged. Climate is shifting. Cheatgrass is invading. Design a restoration plan that gives this ecosystem its best chance — even if 'best chance' means adapting to become something new rather than what it was."*

Watch For

Groups design restoration plans with intervention timelines, species selection, invasive management, and climate adaptation strategies. Present and debate: which plan is most realistic? Most ecologically sound? Present the tension between restoring the past and adapting to the future.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Disturbance Severity, Seed Bank and Propagule Availability, Climate Suitability
- **INTERNAL:** Pioneer Species Establishment, Soil Development Rate, Biodiversity Accumulation, Nutrient Cycling Efficiency, Trophic Web Complexity, Invasive Species Pressure, Recovery Completeness Index

Relationships

- **Disturbance Severity to Soil Development Rate:** NEGATIVE (-) — More severe disturbance destroys more soil, requiring longer soil formation periods. Complete sterilization (volcanic eruption) means soil must form from rock weathering over centuries; moderate disturbance (fire) leaves soil structure partially intact.
- **Seed Bank and Propagule Availability to Pioneer Species Establishment:** POSITIVE (+) — Richer seed banks provide more species and more individual propagules for colonization. When seed banks are destroyed, pioneer establishment depends on long-distance dispersal from surrounding areas, which is slower and less diverse.
- **Pioneer Species Establishment to Soil Development Rate:** POSITIVE (+) — Pioneer species accelerate soil formation through root exudates that weather rock, nitrogen fixation that adds nutrients, and organic matter accumulation from dead plant material. Without pioneers, soil formation proceeds only through physical and chemical weathering.
- **Soil Development Rate to Biodiversity Accumulation:** POSITIVE (+) — Functional soil supports a wider range of plant species through nutrient availability, water retention, and microbial partnerships. As soil deepens and enriches, species that require specific soil conditions can establish.
- **Biodiversity Accumulation to Trophic Web Complexity:** POSITIVE (+) — More plant species support more herbivore species, which support more predator species. Biodiversity at the base of the food web cascades upward through trophic levels, creating the complex food webs that characterize mature ecosystems.
- **Trophic Web Complexity to Nutrient Cycling Efficiency:** POSITIVE (+) — More complex food webs process and recycle nutrients through more pathways, reducing nutrient loss from the ecosystem. Decomposers, detritivores, and mycorrhizal networks all increase nutrient retention as trophic complexity increases.
- **Invasive Species Pressure to Recovery Completeness Index:** NEGATIVE (-) — Invasive species compete with native species for establishment sites, nutrients, and light. When invasives dominate early succession, they can permanently alter the community trajectory and prevent native species from establishing, reducing recovery completeness.
- **Climate Suitability to Recovery Completeness Index:** POSITIVE (+) — When climate conditions support the historical ecosystem, full recovery is possible given sufficient time and biological legacy. When climate has shifted beyond the tolerance of key species, recovery plateaus at a lower completeness level because the original community cannot re-establish.

Key Discoveries

- Soil is the critical bottleneck in ecosystem recovery — with intact soil, secondary succession can restore functional ecosystems in 20-50 years, but without soil, primary succession requires centuries because soil must be built from scratch through slow weathering and biological processes
- The seed bank and propagule availability determine the initial trajectory of recovery — when diverse seed banks survive, recovery proceeds through predictable successional stages, but when seed banks are destroyed, the first species to arrive by long-distance dispersal may not be the optimal community and can lock in suboptimal recovery pathways
- Resilience thresholds exist where recovery becomes impossible — when disturbance severity exceeds the ecosystem's capacity to regenerate and climate conditions have shifted, the system undergoes a regime shift to a fundamentally different stable state rather than recovering to its original condition

- Invasive species exploit the vulnerability window during early succession — disturbed ecosystems with open space, available nutrients, and reduced competition are ideal habitats for aggressive invasive species that can permanently alter the successional trajectory and prevent recovery of native communities

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Ecosystems always recover naturally if left alone"	While many ecosystems have remarkable self-repair capacity, recovery is not automatic or guaranteed. Ecosystems can cross resilience thresholds beyond which the original feedback loops cannot re-establish. Invasive species can permanently alter successional trajectories. Climate change can make the original ecosystem impossible under new conditions. In many cases, active restoration — removing invasives, stabilizing soil, reintroducing key species — is required to guide recovery in a desired direction rather than leaving it to chance.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Succession always follows the same predictable sequence"	While general patterns of succession are predictable (pioneer species first, then competitive species, then climax community), the specific trajectory depends on which species arrive first (priority effects), which seed sources are available, what the soil conditions are, and what the current climate supports. Two areas with identical disturbance can follow very different successional pathways depending on these contingent factors. The concept of a single fixed 'climax community' has been replaced by the recognition that multiple stable states are possible for any given location.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Climate change will not affect ecosystem recovery because species will just adapt"	Most tree species take decades to centuries to reproduce and evolve, while climate is changing over decades. Species cannot adapt fast enough through natural selection to keep pace with the rate of climate change. Instead, species ranges shift — they survive where conditions remain suitable and disappear where conditions have changed. For a recovering ecosystem, this means that the species adapted to the historical climate may not be able to re-establish, and the species adapted to the future climate may not yet be present in the local seed bank. This mismatch is why assisted migration and forward-looking restoration are becoming necessary.